SINGAPORE SCHOOL KINDERLAND

Department of Information Technology & Computer Science · Academic Year 2026–2027

PRIMARY 6 · IT CLASS TEST 2 (TERM 2) · TEACHER ANSWER KEY & MARKING SCHEME

Full Solutions & Evaluation Criteria · Total: 25 Marks

Document: Teacher's Official Marking Scheme	**Grade / Level:** Primary 6 (All Sections)	**Maximum Marks:** 25

SECTION A: MULTIPLE CHOICE QUESTIONS [5 MARKS]

1. **(b) =** (Equal sign starts all formulas) [1 Mark]
2. **(c) D8** (Column letter followed by Row number) [1 Mark]
3. **(c) AVERAGE** (Pre-defined function for arithmetic mean) [1 Mark]
4. **(b) Record** (Horizontal row containing complete tuple of attributes) [1 Mark]
5. **(c) Primary Key** (Unique field identifier strictly forbidding duplicates/nulls) [1 Mark]

SECTION B: FILL IN THE BLANKS [5 MARKS]

1. **Name Box** (or Address Box) [1 Mark]
2. **RDBMS** (Relational Database Management System) [1 Mark]
3. **=COUNT(B1:B10)** [1 Mark]
4. **Field** (or Column / Attribute) [1 Mark]
5. **Query** [1 Mark]

SECTION C: SPREADSHEET ANALYSIS & FORMULA CALCULATION [5 MARKS]

1. Formula in D1: **=B1*C1** (Also accept: **=PRODUCT(B1,C1)**) [1 Mark]
2. Formula in D4: **=SUM(D1:D3)** (Also accept: **=D1+D2+D3**) [1 Mark]
3. Maximum formula: **=MAX(B1:B3)** [1.5 Marks]
4. Error display: ### (Hash symbols indicate insufficient column width) [1.5 Marks]

SECTION D: OPENOFFICE BASE DATABASE DESIGN [5 MARKS]

Data Types:

(a) Book_ID: **Text / Varchar** [0.5 Mark]

(b) Title: **Text / Varchar** [0.5 Mark]

(c) Price: **Number / Decimal / Currency** [0.5 Mark]

(d) Date_Purchased: **Date** [0.5 Mark]

(e) Is_Available: **Yes/No (Boolean)** [0.5 Mark]

Primary Key Selection & Justification: [2.5 Marks]

• Primary Key: **Book_ID** [1 Mark]

• Justification: Every book possesses a distinct, unique identifier code that will never be repeated or left empty, whereas titles, prices, or dates can have duplicate values across multiple books. [1.5 Marks]

SECTION E: DESCRIPTIVE TECHNICAL ANSWERS [5 MARKS]

1. **Calc vs Base Comparison:** [2 Marks]

• *Difference 1:* Calc is designed for mathematical calculations, budgeting, numerical formulas, and graphs, whereas Base is designed to organize, store, and manage large collections of interrelated structured records.

• *Difference 2:* Calc organizes data in flat sheets of rows and columns; Base allows linking multiple tables using relationships (foreign keys) to prevent duplicate data entry.

2. **Forms & Reports in Base:** [2 Marks]

• *Forms:* Provide an interactive, user-friendly graphical interface to enter, view, and edit database records safely without altering table structures directly.

• *Reports:* Formatted, print-ready summaries of database queries designed to present data attractively for administrative or management decision-making.

3. **Data Redundancy:** [1 Mark]

Data redundancy refers to unnecessary repetition of the same data across multiple places. Databases eliminate redundancy by storing data once in related tables and referencing it via Primary Keys.